

Self-contained validation: NOT YET RUN. Before considering this report final, copy config.ini + scripts/ to a scratch location (or point OUTPUT_ROOT at an isolated subdirectory), fill in WORKSPACE_ROOT, and run scripts/run_all.sh end to end — then re-call session_final_report with validated=True.

Megatron-DeepSpeed GPT-3 Medium 350M — dftracer Optimization Pipeline Report

1. Executive Summary

- **Session ID:** megatron_deepspeed/20260726_185517
- **Application:** Megatron-DeepSpeed (argonne-lcf fork), GPT-3 Medium 350M (pretrain_gpt.py)
- **System:** Tuolumne (LLNL), AMD MI300A APUs (unified CPU+GPU HBM), ROCm, Cray PE, Flux scheduler
- **Scope note (MANDATORY):** BERT pretraining was in the original plan and was **deferred and never started**. Every finding below covers GPT-3 Medium 350M only.
- **Pipeline status:** Partial success. Steps 1-9 (setup through 4-dimension optimization) complete and measured. Step 10 (this report) complete but **not validated end-to-end**. Pipeline Policy rule 12 (dftracer_service node counters) was violated for all runs — no HBM/occupancy telemetry exists this session.
- **Top 3 bottlenecks found:** (1) Compute — 67% of wall time in training step; (2) Init/setup — 29% (model build 15.2% + import storm + optimizer-state construction 2.0%); (3) I/O — negligible (0.8%, zero POSIX ops during steady-state training).
- **Top 3 optimizations applied (measured):** NONE beat baseline. RCCL/ncccl backend swap (+1.2%, noise), DeepSpeed ZeRO-1 (-20.2%, regression), bf16 (~-25%, regression) were all applied and measured; none were kept.
- **Key metric:** Baseline steady-state throughput = **145.4 samples/sec** (iters 2-3, iter 1 excluded as apex-JIT warmup).
Headline finding: the baseline configuration is already near-optimal for this model size on this stack — every lever tested was neutral or a regression.

2. Baseline Detail

Run	Config	Measured metric	Used to compare against
baseline (16-rank, verified genuine)	4 nodes x 4 ranks MI300A, DeepSpeed ZeRO-0, fp16, gloo backend, seq_len 2048, micro_batch 4, global_batch 256, 3 train iters, 4000-doc pre-sliced oscar/openwebtext GPT2BPE	Loss 11.0161 -> 10.1995 -> 9.8462. Steady-state 145.4 samples/sec (iter2=1.764s, iter3=1.760s; iter1=24.53s JIT warmup excluded). 16/16 ranks non-empty traces, ~263,418 events.	Reference for all 4 optimizer-dimension variants
opt_compute/base20* (compute-agent's own baseline, 4 reps)	Same config, 20 train iters	142.5 sps median, within-run CV <1.3%	Reference for compute agent's bf16/hipblaslt/rocblas arms

Only one true baseline exists (the compute agent's base20 reps re-measure the same configuration with more iterations/replicates for statistical power, not a distinct baseline point). No re-baselining occurred this session — the 16-rank baseline required ~10 attempts to obtain a genuine distributed run, but every earlier attempt is archived as invalid and explicitly excluded from all reported numbers.

Archived-invalid data (MANDATORY EXCLUSION): baseline/traces/raw_prev/ and baseline/traces/compact_prev/ contain traces from earlier failed baseline attempts and are NOT part of any reported metric.

3. Per-Agent Contributions

dftracer-session-setup / system-detect

- Detected Tuolumne system config: MI300A ROCm, Cray PE, Flux. Artifacts: `session_report.md`, `artifacts/02_session_detect.log`

dftracer-build-app

- Built Megatron-DeepSpeed against a ROCm PyTorch wheel (CUDA wheel resolves by default and had to be overridden). Artifacts: `artifacts/03_build_app.log`, `artifacts/03_session_configure.log`, `build/`

dftracer-build-dftracer

- Installed dftracer in FUNCTION mode (Pipeline Policy rule 13), linked `libdftracer_core.so`. Artifacts: `artifacts/04_install_dftracer.log`

dftracer-annotator

- Annotated 202/207 functions = **97.6% coverage** across 10 files: `pretrain_gpt.py`, `megatron/{training,checkpointing,utils,initialize,arguments}.py`, `megatron/data/{gpt_dataset,indexed_dataset,data_samplers,blendable_dataset}.py`.

dftracer-build-smoke

- Single-GPU smoke test passed after ~7 attempts. Artifacts: `artifacts/05_session_run_smoke_test.log`, `artifacts/06_build_smoke.log`, `artifacts/step6_attempt7_smoke*.log`

dftracer-tracer (baseline, ~10 attempts, every failure a distinct real bug)

- Final genuine 16-rank distributed baseline: 4 nodes x 4 ranks, non-empty traces on all 16 ranks, ~263,418 events.
- **Violated Pipeline Policy rule 12:** no `session_service_start/stop` bracketing applied to any run — no `service_*.pfw` node counters exist for baseline or any variant.
- Artifacts: `baseline/record/`, `artifacts/baseline_run_FINAL.log`, `artifacts/07_baseline_run.log`, plus 9 archived attempt logs kept for provenance only.

dftracer-analyzer

- dfanalyzer generic preset, 32 workers, over the full baseline trace set (489.746s aggregate / 263,418 ops). Artifacts: `artifacts/analyze_baseline.log`, `analyzer_checkpoint/`

dftracer-diagnoser

- Classified the workload as **compute-bound** (67% training step) with negligible, structurally-explainable I/O.

dftracer-optimizer (all 4 dimensions dispatched per Pipeline Policy rule 14)

- `dftracer-optimizer-compute`: RCCL backend, ZeRO-1, bf16, HIPBLASLT flag — all measured, all neutral-or-regression.
- `dftracer-optimizer-io`: full 12-item checklist walked; verdict "not applicable" (zero steady-state POSIX ops).
- `dftracer-optimizer-memory`: optimizer-state construction measured at 2.0% of pretrain span — correctly predicted ZeRO-1 would not help.

- dftracer-optimizer-communication: RCCL/gloo swap measured at +1.2% (a trace-derived 35% estimate was ~30x too high — finding #1).
- Artifacts: opt_zero1/, opt_nccl/, opt_bf16/, opt_bf16b/,
opt_compute/{base20, base20_rep1-4, bf16, bf16_rep1-2, hipblaslt, hipblaslt_rep1-3, rocblas, rocblas}
artifacts/09_optimizer_compute_*.log, artifacts/run_opt_*.log

4. Tool Findings vs Manual Analysis

Finding	Source	Details
Training=67%, init=29%, POSIX=0.8% wall-time split	TOOL	dfanalyzer generic preset, 32 workers, 489.746s / 263,418 ops
Steady-state training issues zero POSIX ops	TOOL	dftracer trace query, iters 2-3 only
106,392 STDIO ops are ~99.5% Python import storm (36,894 fclose + 28,353 fopen64 + 15,407 feof)	MANUAL	Custom trace-event breakdown by call site — dfanalyzer's summary does not separate import traffic from application STDIO
get_model=24.807s (15.2% of rank-0 pretrain span) vs optimizer-state construction=3.26s (2.0%)	TOOL	dfdiagnoser / dfanalyzer per-function timing
apex fusion status (FusedAdam, amp_C, fused_layer_norm_cuda, fused_weight_gradient_mlp_cuda)	MANUAL	Direct Python import probe on a compute node — corrects an earlier wrong inference from DeepSpeed's op-builder [NO]/[OKAY] table
RCCL backend swap measured delta (+1.2%) vs gloo baseline	TOOL	comparator + direct wall-clock throughput comparison
bf16 regression (~-25%), ZeRO-1 regression (-20.2%)	TOOL/MANUAL	Wall-clock throughput from run logs; comparator run on trace deltas where traces existed
flash-attn flags silently no-op	MANUAL	Source inspection of DeepSpeed 0.19.3 FlashAttentionBuilder + Megatron arg handling

5. Bottleneck Diagnosis

Rank	Bottleneck	Severity	Evidence
1	Compute (training step GEMMs)	Dominant (67% wall time)	dfanalyzer: training 328.909s of 489.746s aggregate
2	Init/setup (model build + optimizer state + import storm)	Secondary (29%)	init 140.713s; get_model 24.807s (15.2%); optimizer-state 3.26s (2.0%)
3	Communication (grad all-reduce)	Minor, smaller than trace-based estimate predicted	RCCL swap measured only +1.2% vs a ~35% trace-derived estimate

4	I/O	Negligible / structurally absent in steady state	POSIX 3.788s (0.8%, 10,233.75 MB @ 2,701.95 MB/s), all in iteration 1; zero POSIX ops in iters 2-3
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6. Optimization Proposals (all 4 dimensions walked per rule 14)

Compute (L1/L2)

- **RCCL/nccl distributed backend** (vs. gloo). Citation: Thakur, Rabenseifner, Gropp, "Optimization of Collective Communication Operations in MPICH," IJHPCA 2005. Expected: faster grad all-reduce. **Status: applied, measured +1.2% (noise) — not credited, though retained as safe default.**
- **DeepSpeed ZeRO stage 1.** Citation: Rajbhandari et al., "ZeRO: Memory Optimizations Toward Training Trillion Parameter Models," SC 2020. **Status: applied, measured -20.2% — REGRESSION, reverted.**
- **bf16 instead of fp16.** Citation: Kalamkar et al., "A Study of BFLOAT16 for Deep Learning Training," arXiv:1905.12322. **Status: applied, measured ~-25% — REGRESSION, reverted.**
- **TORCH_BLAS_PREFER_HIPBLASLT=1/0.** Citation: AMD ROCm hipBLASLt documentation. **Status: applied both directions, -2.0% / -0.6% — inert both ways, confirms dead knob.**
- **FlashAttention.** Citation: Dao et al., "FlashAttention," NeurIPS 2022, <https://arxiv.org/abs/2205.14135>. Estimated ~17% (highest-ceiling untested lever). **Status: NOT measured — flash_attn absent, DeepSpeed 0.19.3 FlashAttentionBuilder returns None, Megatron silently falls back to use_flash_attn=False. Needs a multi-hour ROCm/CK build; deferred.**

I/O (full checklist, all "not applicable")

- All 12 checklist items walked and marked **not applicable**: steady-state training issues zero POSIX ops; the only I/O (10.2 GB) occurs in iteration 1 and iterations 2-3 run entirely from page cache.
- **num_workers=0**: applicable candidate, value purely a production-scale argument. **Status: applicable, NOT measured at this scale.**

Memory (L1/L2/L3)

- **ZeRO-1 optimizer-state partitioning** (same proposal as Compute) — optimizer-state construction measured at only 2.0% of pretrain span, correctly predicting no capacity pressure to relieve on MI300A's unified HBM. **Status: applied, measured, regression, confirms the memory-dimension prediction.**
- Checkpoint memory/I-O: **not applicable/unmeasured** — zero checkpoints written (save-interval 10 > 3 train iterations).

Communication (L1/L2/L3)

- **RCCL/nccl backend** — **Status: applied, measured +1.2%, NO CHANGE within noise** (session's most counter-intuitive result, see finding #1).
- Gradient-accumulation-fusion/overlap tuning: not separately tested; grad-accum=4 over 16 ranks already keeps all-reduce a small fraction of iteration time per the RCCL-swap result.

7. Optimization Ledger

#	Dimension	Level	Change	Applied?	Measured delta	Significance/replicat	Verdict
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1	communication	L2	gloo -> nccl (RCCL) distributed backend	yes	+1.2% (147.1 vs 145.4 sps)	1 replicate	harmless-but-not-credible (within noise; adopt as safe default)
2	memory	L2	DeepSpeed ZeRO stage 1	yes	-20.2% (116.0 vs 145.4 sps)	1 replicate	reverted (regression)
3	compute	L1/L2	fp16 -> bf16	yes	~-25% (106.1 mine / 107.5 compute-agent, 2 reps)	1 + 2 reps	reverted (regression, both agree)
4	compute	L2 (env var)	TORCH_BLAS_PREFER_HIPBLAS=1	yes	-20%	3 reps	reverted (small regression)
5	compute	L2 (env var)	TORCH_BLAS_PREFER_HIPBLAS=0 (control)	yes	BLASLT=0 -0.6%	3 reps	harmless-but-not-credible (both directions inert -> knob dead)
6	compute	L1	FlashAttention	no (build blocker)	not measured	0	needs a ROCm/CK build — highest-ceiling untested lever (~17% est.)
7	io	L1-L3 (12-item checklist)	(all 12 strategies)	no	not measured (steady-state I/O = 0 ops)	n/a	not-applicable — structurally absent bottleneck
8	io	L1	num_workers>0 / prefetch_factor	no (scale-gated)	not measured	n/a	applicable-not-measured — needs production-scale dataset
9	memory	n/a	checkpoint I/O tuning	no	not measured (zero checkpoints written)	n/a	not-applicable this session, unmeasured

Mechanism paragraphs:

- Row 1 (RCCL swap, +1.2% vs ~35% trace-derived estimate):** At 350M params with global_batch 256 / micro_batch 4 (grad-accum 4) across 16 ranks, the gradient all-reduce payload is small relative to per-rank compute time (~1.76s/step). A trace-derived estimate assuming proportional communication cost predicted ~35% improvement; measured effect was ~30x smaller — a clear demonstration that a plausible trace-derived projection

can be badly wrong when compute/communication overlap is already present. Measure, don't extrapolate.

- Row 2/3 (ZeRO-1 and bf16 regressions):** Both target relieving pressure that doesn't exist at this scale on MI300A's unified HBM — optimizer-state construction is only 2.0% of pretrain time, so partitioning it adds coordination overhead without removing a real constraint; bf16's regression is consistent with apex's fused kernels already being fp16-tuned (finding #2), losing a well-optimized fast path rather than gaining one.
- Row 5 (hipBLASLt env var, inert both directions):** Regresses in BOTH "on" and "off" (control) positions relative to baseline, by amounts smaller than cross-replicate noise — not a real lever on this stack; treat as absent from the tuning surface.

7b. N-Way Comparator Table

Config	Level	Scale	Throughput (samples/sec)	Δ vs baseline	Replicates	Range/CV	Overlaps baseline?	Verdict
baseline (gloo, ZeRO-0, fp16)	—	4N x 4 ranks	145.4 (mine) / 142.5 (compute agent)	reference	1 + 4	compute-agent CV <1.3%	—	kept (final config)
nccl/RCCL backend	L2 comm	4N x 4 ranks	147.1	+1.2%	1	not established (single rep)	overlaps (within ~4% cross-replicate noise)	rejected-no-effect (safe default only)
DeepSpeed ZeRO-1	L2 memory	4N x 4 ranks	116.0	-20.2%	1	not established (single rep)	does not overlap	rejected-regression
bf16 (mine)	L1/L2 compute	4N x 4 ranks	106.1	-27.0%	1	n/a	does not overlap	rejected-regression
bf16 (compute agent)	L1/L2 compute	4N x 4 ranks	107.5	-26.1%	2	n/a (2 reps)	does not overlap	rejected-regression (corroborates row above)
TORCH_BLASLT=1	PREFER_HIPBLASLT=1	4N x 4 ranks (20-iter base)	~-2.0% relative	-2.0%	3	within noise band	overlaps	rejected-no-effect
TORCH_BLASLT=0 (control)	PREFER_HIPBLASLT=0 L2 env	4N x 4 ranks	~-0.6% relative	-0.6%	3	within noise band	overlaps	rejected-no-effect (control confirms dead knob)

Footnote: the compute agent's base20/variant reps ran 20 training iterations (vs. 3 for baseline/nccl/zero1/bf16 runs); both exclude iteration 1 as apex-JIT warmup, but absolute sample counts differ, so cross-series percentages are reported separately (mine vs compute-agent), not pooled. All rows share the 4-node x 4-rank / MI300A / ROCm / fp16-unless-noted configuration.

Code/config snippets:

nccl backend — run_opt_nccl.sh:

```
- --distributed-backend gloo + --distributed-backend nccl
```

ZeRO-1 — ds_config_zero1.json vs ds_config_baseline.json:

```
- "zero_optimization": { "stage": 0 } + "zero_optimization": { "stage": 1 }
```

bf16 — ds_config_bf16.json / run_opt_bf16.sh / run_opt_bf16b.sh:

```
- "fp16": { "enabled": true } + "bf16": { "enabled": true }
```

plus --fp16 -> --bf16 on the Megatron CLI args.

hipBLASLt — opt_launch_hipblaslt.sh / opt_launch_norocblas.sh:

```
export TORCH_BLAS_PREFER_HIPBLASLT=1 # variant export TORCH_BLAS_PREFER_HIPBLASLT=0 # control
```

8. Iteration Results

Iteration	Applied change	Metric before	Metric after	Delta%
opt_nccl	gloo -> nccl backend	145.4 sps	147.1 sps	+1.2%
opt_zero1	ZeRO-0 -> ZeRO-1	145.4 sps	116.0 sps	-20.2%
opt_bf16 / opt_bf16b	fp16 -> bf16	145.4 sps (mine) / 142.5 sps (compute agent)	106.1 sps / 107.5 sps	-27.0% / -26.1%
opt_compute/hipblaslt*	HIPBLASLT=1	142.5 sps	~139.6 sps	-2.0%
opt_compute/rocblas*	HIPBLASLT=0 (control)	142.5 sps	~141.7 sps	-0.6%

Best configuration found: the original baseline (gloo, ZeRO-0, fp16, default hipBLASLt state) — no tested variant beat it. **Convergence status: exhausted** — all four optimizer dimensions were walked per rule 14 and every measured lever was neutral-or-regressive; the loop ran its full planned scope and the conclusion is that baseline is the best config found.

9. Lessons Learned

- New lessons persisted this session (see git-tracked memory under src/dftracer_agents/.agents/workspace/memory/, listed in Finalization): Cray LD_LIBRARY_PATH ordering for Fortran runtime, @dft.log/@classmethod decorator ordering, mpi4py Cray soname mismatch, Tuolumne ROCm torch wheel requirement, Tuolumne ROCM_HOME auto-detection gap for DeepSpeed op-builder, Megatron-DeepSpeed JIT/lock pitfalls, launcher world-size-1 trap, smoke-test data-slicing, ROCm apex/run pitfalls, torchvision ABI mismatch.
- Pitfall discovered and corrected mid-session: a false premise that ROCm apex produced only a Python fallback (no fused HIP kernels), based on misreading DeepSpeed's op-builder table. Corrected by direct import probe (finding #2); preserved in the report rather than silently dropped.
- No new MCP tool changes were required; findings were produced via existing analyze/diagnose/comparator tools plus targeted manual trace-event breakdowns where the tool's aggregate summary didn't separate import-storm STDIO from application STDIO.

10. Recommendations for Future Runs

- **What worked well:** the 4-node/16-rank baseline, once genuinely distributed (attempt 10), gave a clean, reproducible 2-iteration steady-state signal; the full 4-dimension optimizer dispatch surfaced the counter-intuitive RCCL finding a narrower optimizer would likely have missed.
- **What needs improvement:** (1) bracket every run with `dftracer_service` node-counter start/stop per rule 12 — this session did not, and the memory dimension had zero HBM/occupancy data as a result; (2) avoid launching competing jobs on shared allocations — two of this session's own launches collided with the compute agent's replicate study, silently SIGABRT-killing one bf16 replicate.
- **Known limitations:** all findings scoped to a 4000-doc pre-sliced dataset at 3-20 iterations; I/O-negligibility and dataloader-stall conclusions do not generalize to production-scale corpora or longer runs; FlashAttention remains the highest-ceiling untested lever pending a ROCm/CK build.

11. Remaining Work / Resume Point

Outstanding/incomplete items:

- FlashAttention build + measurement — highest-ceiling untested lever (~17% est.), blocked on a multi-hour ROCm/CK build not attempted this session.
- `num_workers>0/prefetch` tuning — applicable but unmeasurable at 4000-doc scale.
- Checkpoint I/O performance — entirely unmeasured (zero checkpoints written).
- `dftracer_service` node-counter bracketing (rule 12) — never applied; future runs should add `session_service_start/stop` around every launch.
- BERT pretraining path — never started this session; needs its own STEP 1-9 pass with a BERT-specific config.

Unmeasured but applicable candidates (exact next steps):

- FlashAttention: build ROCm/CK-compatible kernels, re-run with `--use-flash-attn`, compare against the 145.4 sps baseline.
- DataLoader overlap: re-run at production dataset scale with `--num-workers 4` vs `--num-workers 0`.
- **Known tool/skill gaps hit this session, not fixed at the tool level:** none beyond the manual STUDIO-breakdown workaround in Section 4 (a possible future MCP enhancement, not implemented this session).
- **Exact resume instructions:** `run_id = megatron_deepspeed/20260726_185517`. To resume: (1) `session_status(run_id=...)`, (2) `session_read_file(run_id=..., filepath="pipeline_plan.md")`, (3) `memory_read` the `project-megatron-deepspeed-gpt-pipeline` entry, (4) re-invoke `dftracer-optimizer-compute` (STEP 9, FlashAttention sub-task) or a fresh STEP 1 pass scoped to BERT if prioritized.